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(54) **RECEIVER WITH FLEXIBLE LINK SYNCHRONIZATION**

(71) Applicant: **TEXAS INSTRUMENTS INCORPORATED**, Dallas, TX (US)

(72) Inventors: **Jaiganesh Balakrishnan**, Bengaluru (IN); **Kandalla Krishna**, Bengaluru (IN); **Aravind Vijayakumar**, Bengaluru (IN); **Goutham Ramesh**, Bengaluru (IN)

(73) Assignee: **TEXAS INSTRUMENTS INCORPORATED**, Dallas, TX (US)

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H04L 7/00 (2006.01)

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CPC **H04L 7/04** (2013.01); **H04L 1/0057** (2013.01); **H04L 7/0079** (2013.01)

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CPC H04L 7/042; H04L 7/04; H04L 7/0079; H04L 1/0057; H04L 1/0061
See application file for complete search history.

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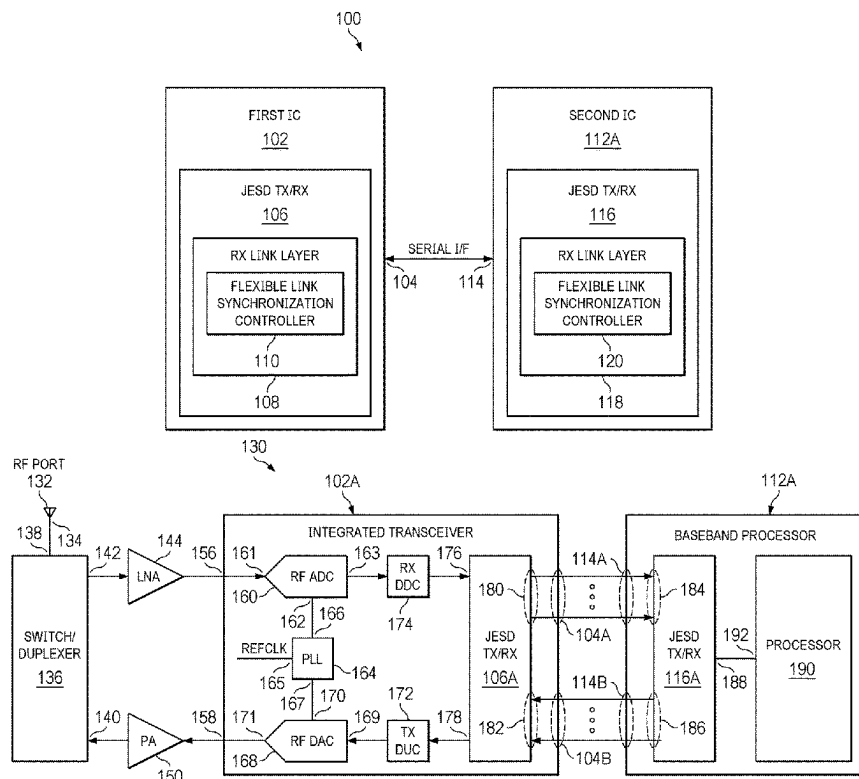
Primary Examiner — Khanh C Tran

(74) *Attorney, Agent, or Firm* — Mandy Barsilai Fernandez; Frank D. Cimino

(57) **ABSTRACT**

A receiver includes: a PHY layer, and a processor coupled to the PHY layer. The processor is configured to: receive a set of data bits from the PHY layer; compare the set of data bits to a sync header pattern; determine a mismatch metric responsive to the comparison and to an adjustable scaling factor, and execute link synchronization operations based on the mismatch metric.

20 Claims, 11 Drawing Sheets



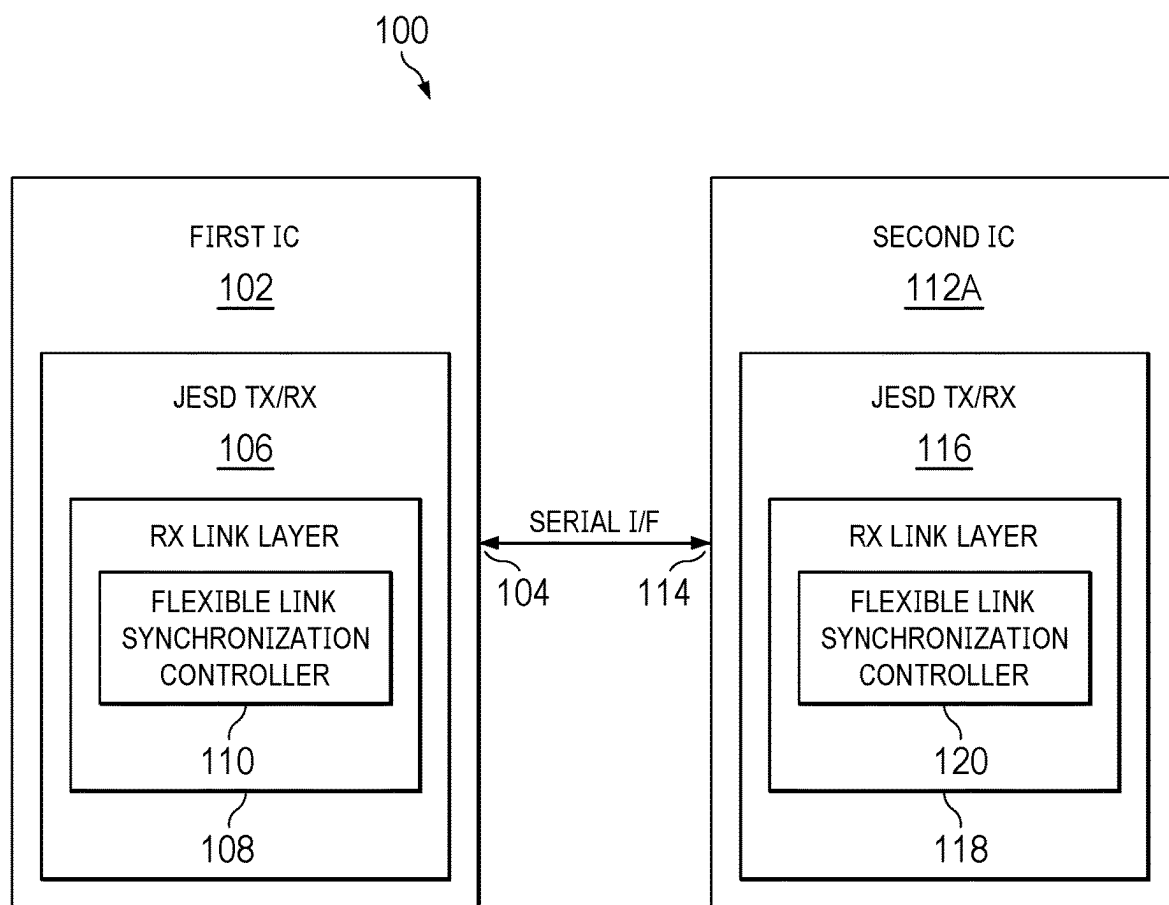


FIG. 1A

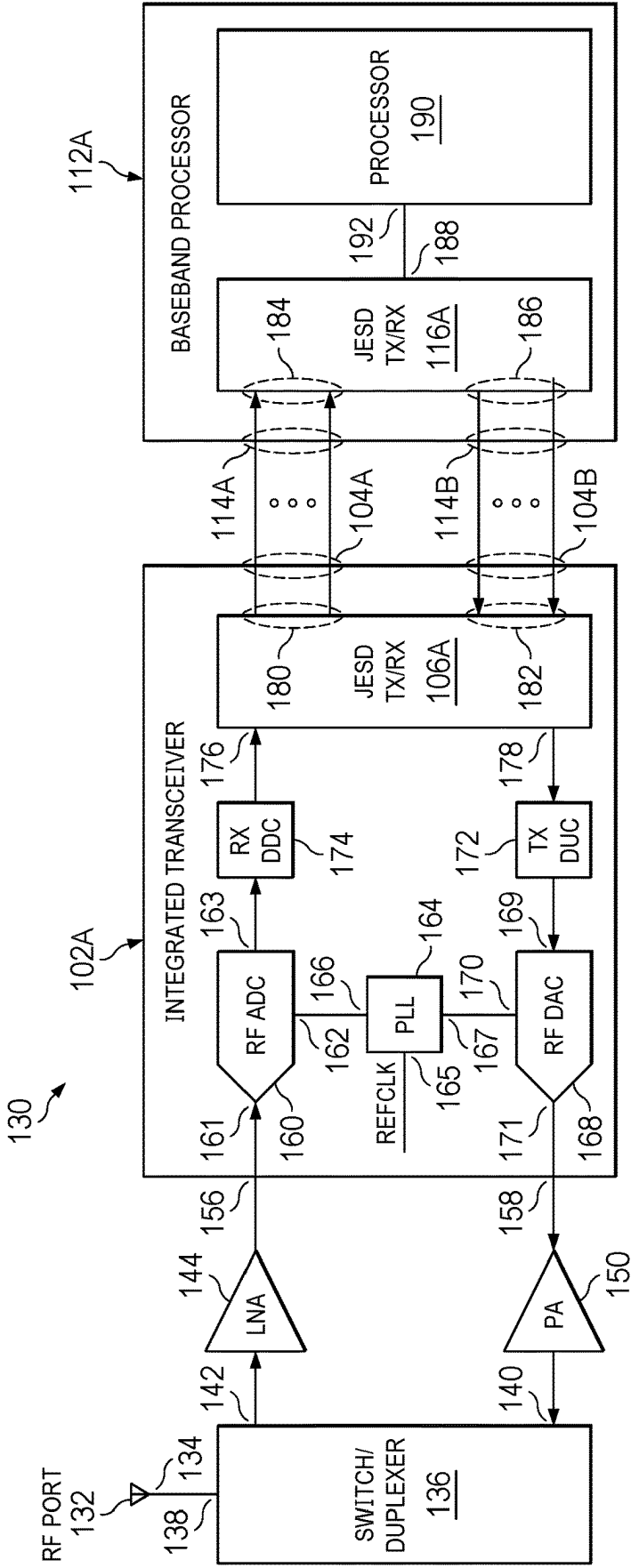


FIG. 1B

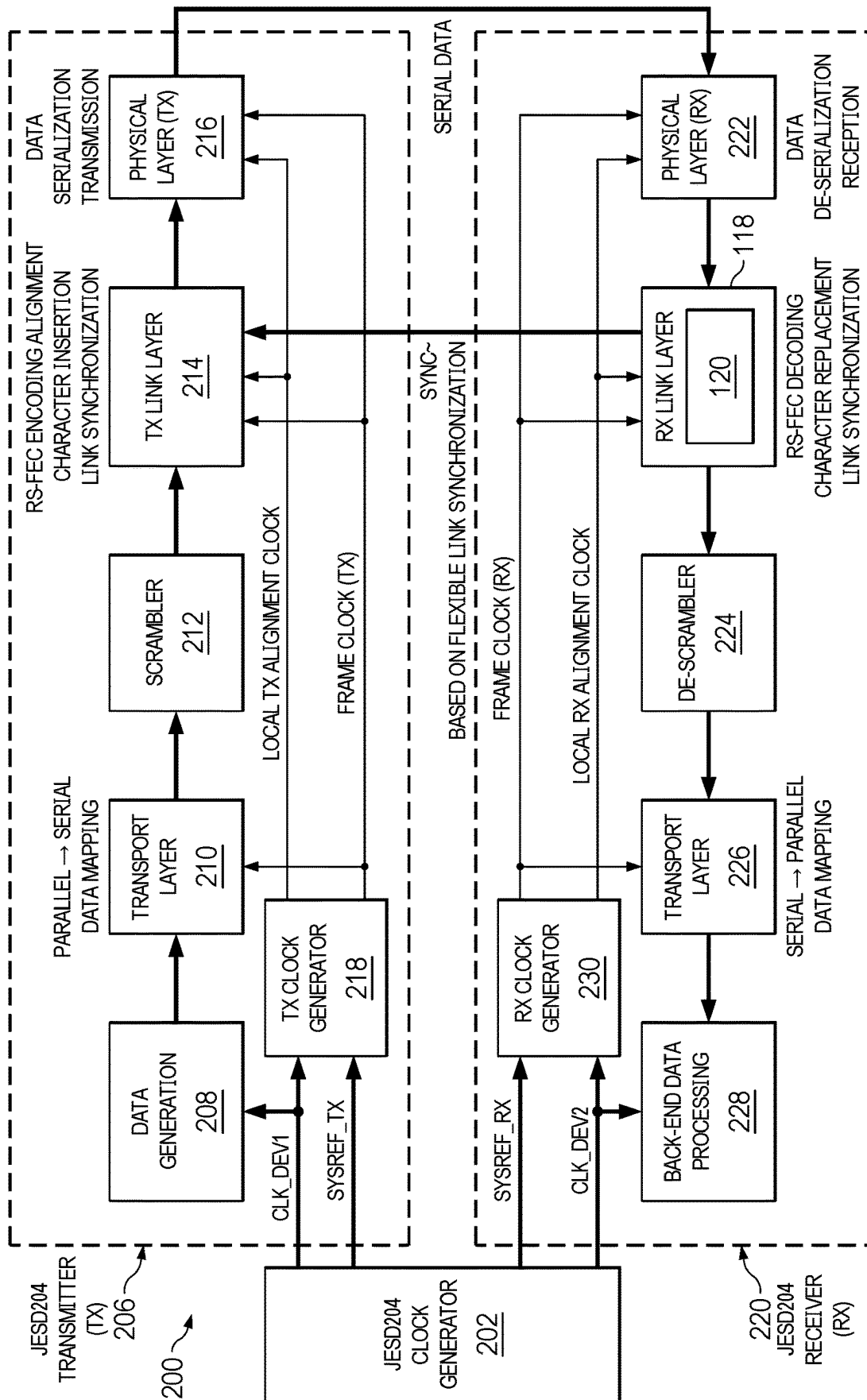


FIG. 2

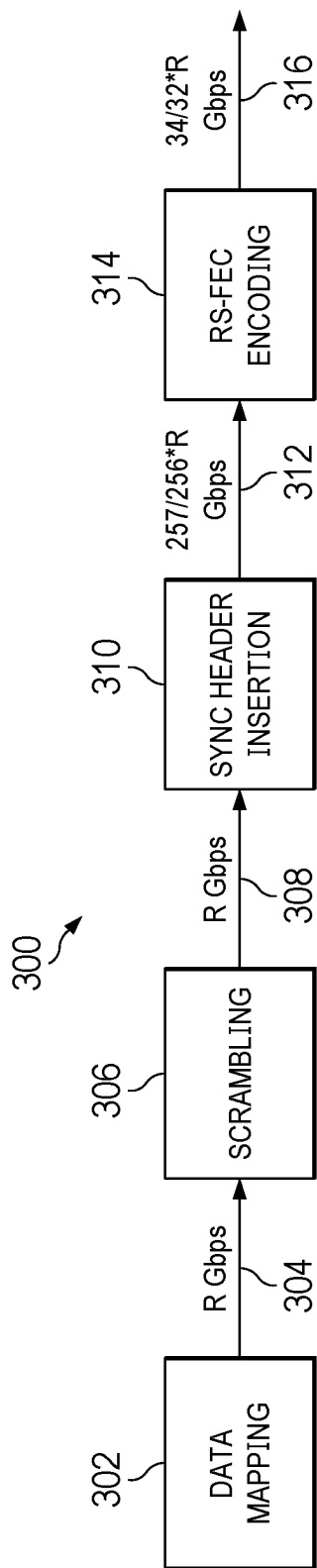


FIG. 3

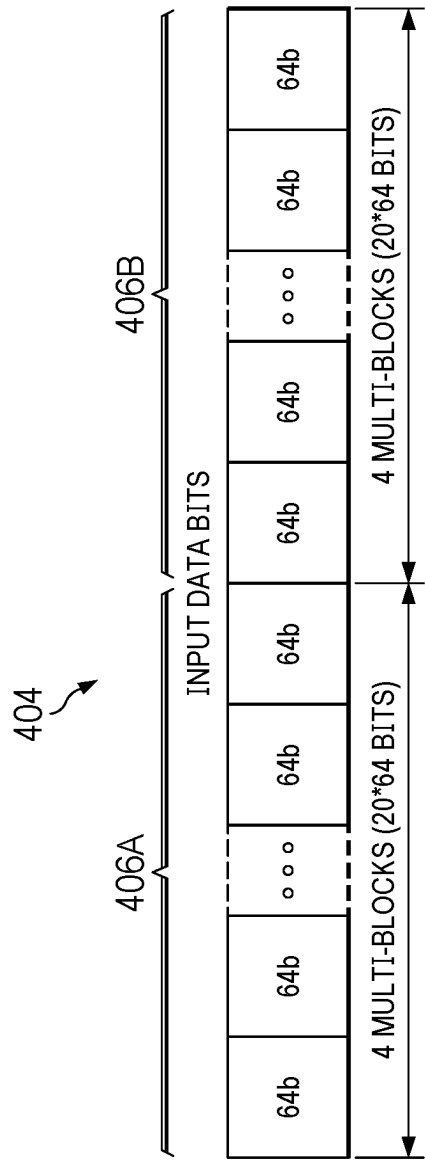


FIG. 4A

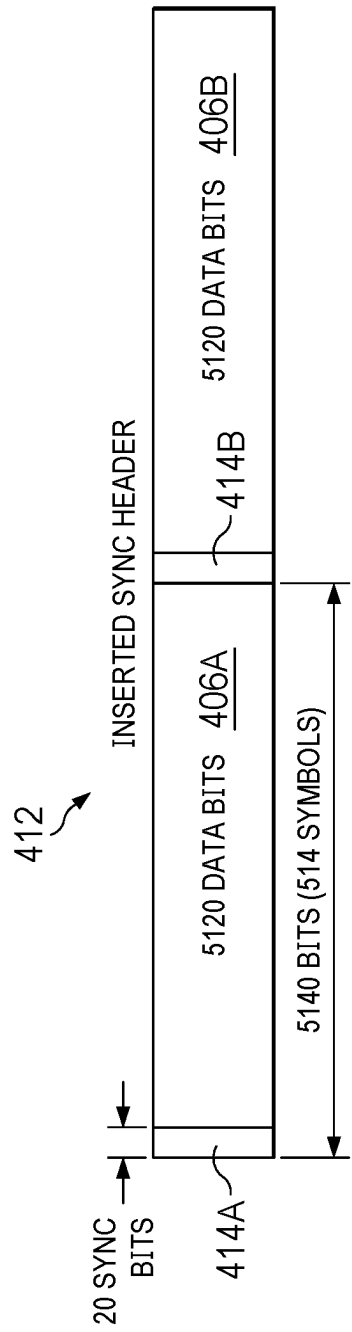


FIG. 4B

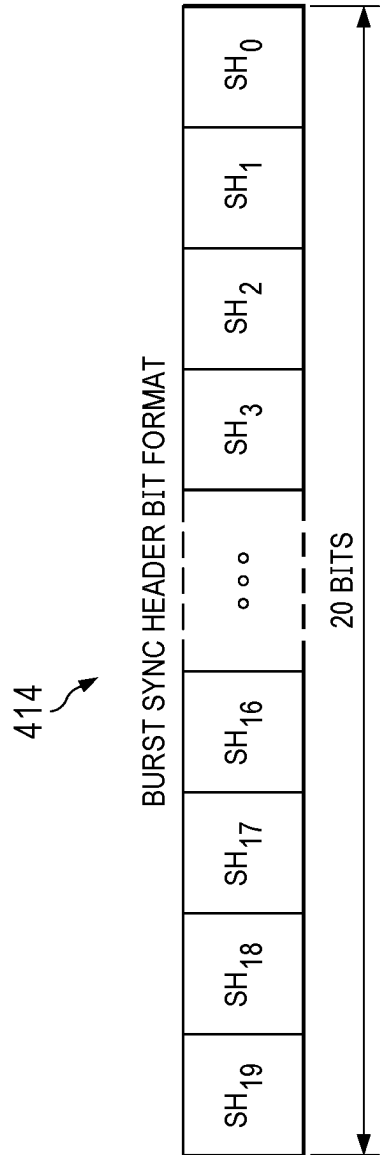


FIG. 4C

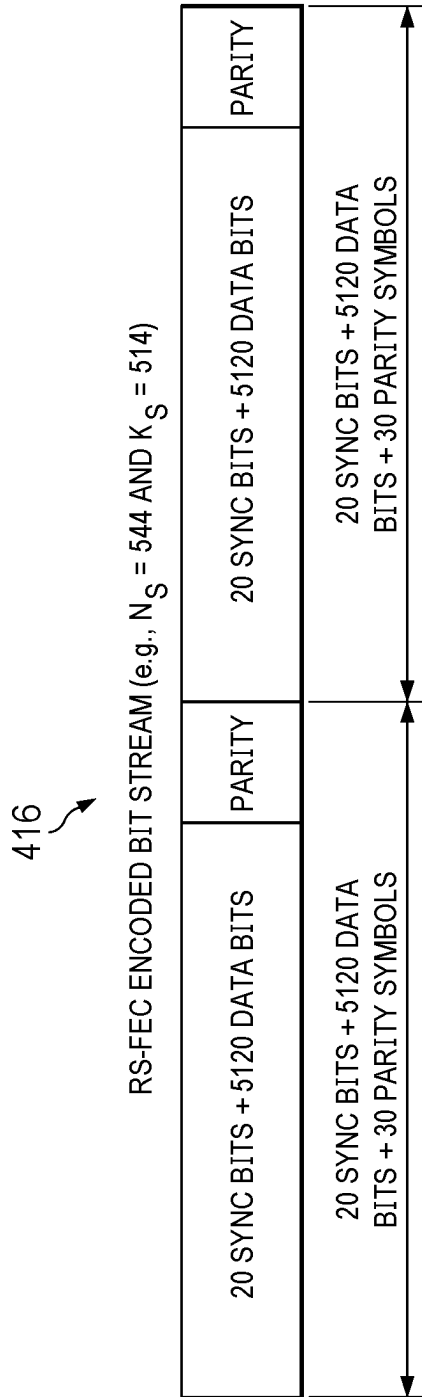


FIG. 4D

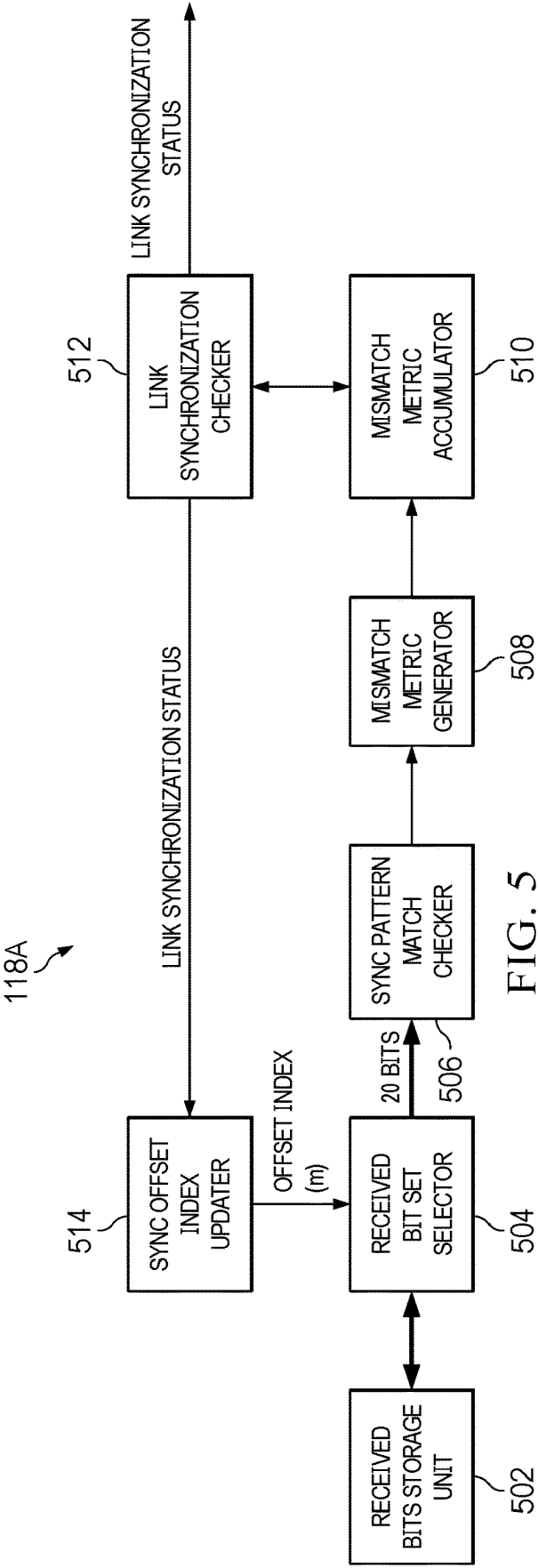


FIG. 5

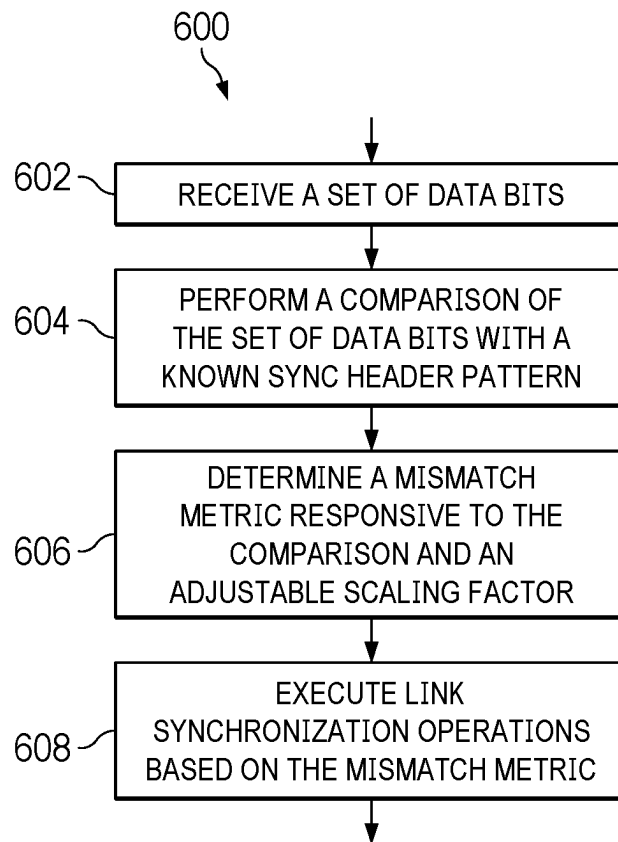


FIG. 6

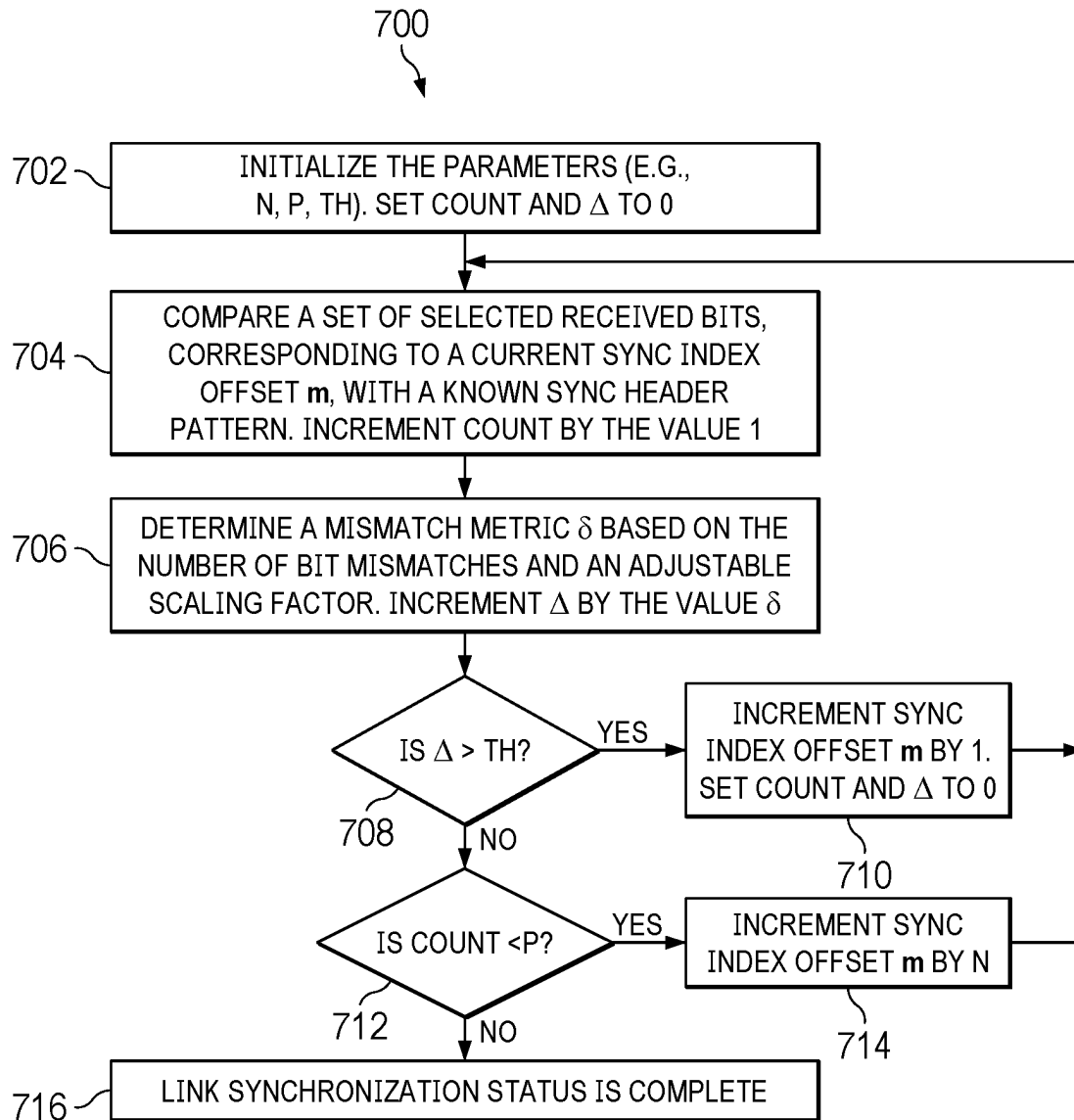


FIG. 7

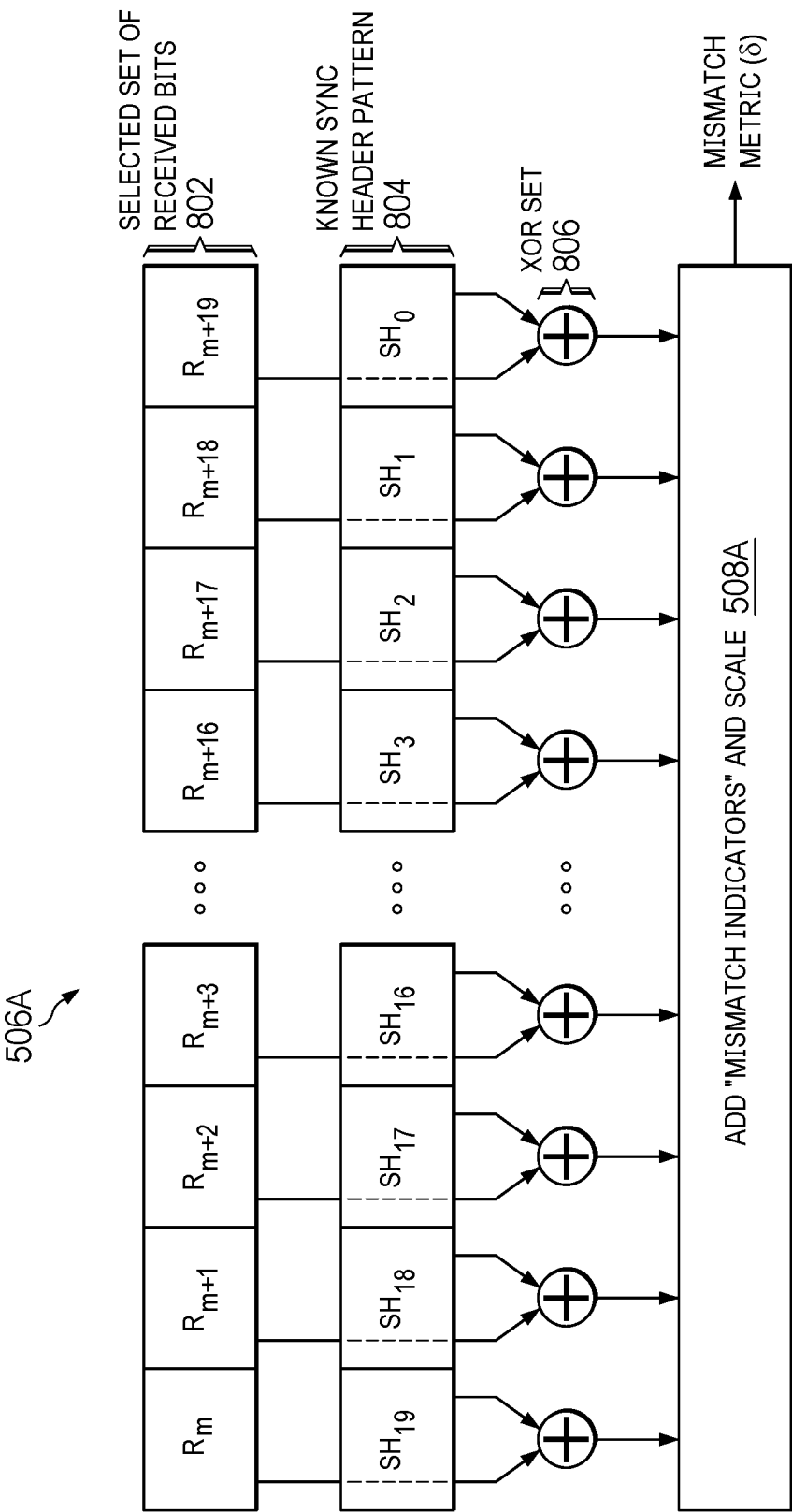


FIG. 8

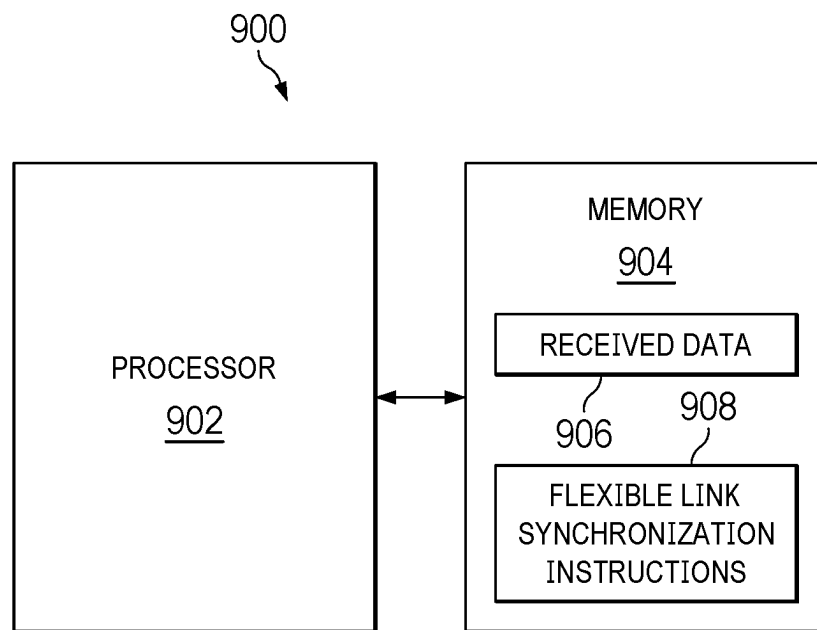


FIG. 9

RECEIVER WITH FLEXIBLE LINK SYNCHRONIZATION

BACKGROUND

Many electronic systems and devices include integrated circuits (ICs) and related communication interfaces. ICs may communicate, for example, via serial communication interfaces and parallel communication interfaces. There are many communication interface protocols, and these communication interface protocols vary with regard to security, data rate, bit error rate, and/or other parameters. JESD is a serial interface standard to transfer data between data converter ICs and logic device ICs. Example data converter IC include analog-to-digital converters (ADCs), digital-to-analog converters (DACs), and/or integrated transceivers with ADCs and DACs. Example logic device ICs include baseband processor ICs, field-programmable gate arrays (FPGAs), and application-specific integrated circuits (ASICs).

JESD currently has two protocol variants: JESD 204B and JESD 204C. JESD 204C supports interface rates up to 32.5 Gbps using non-return-to-zero (NRZ) encoding/decoding in the PHY layer. The bit error rate for NRZ is approximately 10^{-15} . A higher speed variant, JESD 204D, is under development. JESD 204D will support interface rates up to 112 gigabits per second (Gbps) using pulse amplitude modulation 4-level (PAM4) encoding/decoding in the PHY layer. The bit error rate for PAM4 is approximately 10^{-4} to 10^{-5} . To account for the higher BER of PAM4 relative to the BER of NRZ, JESD 204D will employ forward error correction (FEC).

JESD 204D will support higher interface rates and lower interface power consumption and area as fewer lanes are used to transfer data for the given signal bandwidth. For example, an integrated transceiver may use only 16 JESD 204D lanes instead of 32 JESD 204C lanes, to support 16 complex baseband I/Q streams with a 1200 MHz bandwidth.

SUMMARY

In an example, a receiver comprises: a PHY layer; and a processor coupled to the PHY layer. The processor is configured to: receive a set of data bits from the PHY layer; perform a comparison of the set of data bits to a sync header pattern; determine a mismatch metric responsive to the comparison and to an adjustable scaling factor; and execute link synchronization operations based on the mismatch metric.

In another example, a receiver method comprises: receiving a set of data bits; compare the set of data bits to a sync header pattern; determining a mismatch metric responsive to the comparison and to an adjustable scaling factor; and executing link synchronization operations based on the mismatch metric.

In yet another example, a system comprises: a first integrated circuit (IC) having a transmitter; a serial communication interface; and a second IC having a receiver in communication with the transmitter via the serial communication interface. The receiver is configured to: receive a set of data bits; compare the set of data bits to a sync header pattern; determine a mismatch metric responsive to the comparison and to an adjustable scaling factor; and execute link synchronization operations based on the mismatch metric.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a block diagram showing an example system. FIG. 1B is a block diagram showing another example system.

FIG. 2 is a block diagram showing an example JESD204 system.

FIG. 3 is a block diagram showing an example data preparation process.

FIGS. 4A-4D are block diagrams showing example data formats related to the data preparation process of FIG. 3.

FIG. 5 is a block diagram showing an example link synchronization controller.

FIG. 6 is a flowchart showing an example link synchronization method.

FIG. 7 is a flowchart showing another example link synchronization method.

FIG. 8 is a diagram showing an example sync pattern match checker and mismatch metric generator.

FIG. 9 is a block diagram showing an example processing system to perform link synchronization operations.

DETAILED DESCRIPTION

The same reference numbers or other reference designators are used in the drawings to designate the same or similar features. Such features may be the same or similar either by function and/or structure.

FIG. 1A is a block diagram showing an example system **100**. The system **100** may be, for example, a cellular base station system or a radar system. As shown, the system includes a first integrated circuit (IC) **102** and a second IC **112**. In some examples, the first IC **102** includes an integrated transceiver and/or other circuitry having one or more data converters. One example data converter is an analog-to-digital converter (ADC). Another example data converter is a digital-to-analog converter (DAC). In some examples, the second IC **112** includes a baseband processor. One example baseband processor is a field-programmable gate array (FPGA). Another example baseband processor is an application-specific integrated circuits (ASIC). In the example of FIG. 1A, the first IC **102** includes a serial interface **104** and a JESD transceiver (TX/RX) **106**. The JESD transceiver **106** includes an RX link layer **108**. The RX link layer **108** includes a flexible link synchronization controller **110**. The second IC **112** includes a serial interface **114** and a JESD transceiver **116**. The JESD transceiver **116** includes an RX link layer **118**. The RX link layer **118** includes a flexible link synchronization controller **120**.

In some examples, the first IC **102** includes only a JESD transmitter or only a JESD receiver rather than a transceiver. Similarly, the second IC **112** may include only a JESD transmitter or only a JESD receiver in some examples. As another option, the first IC **102** may include the JESD transceiver **106**, while the second IC **112** includes only a JESD transmitter or only a JESD receiver. As another option, the second IC **112** may include the JESD transceiver **116**, while the first IC **102** includes only a JESD transmitter or only a JESD receiver.

Regardless of the particular arrangement, the system **100** includes at least one receiver link layer having a flexible link synchronization controller such as the flexible link synchronization controller **110** of the first IC **102** or the flexible link synchronization controller **120** of the second IC **112**. In some examples, each of the flexible link synchronization controllers **110** and **120** performs flexible link synchronization operations based on a mismatch metric and an adjust-

able scaling factor. The flexible link synchronization operations improve link synchronization when bit error rates (BERs) are above a threshold. While designed for use with the BER of pulse amplitude modulation 4-level (PAM4) encoding/decoding used for JESD 204D, the flexible link synchronization controllers **110** and **120** and related operations may be used with other communication protocols.

FIG. 1B is a block diagram showing another example system **130**. As shown, the system **130** includes an integrated transceiver IC **102A** and a baseband processor IC **112A**. The integrated transceiver IC **102A** is an example of the first IC **102** in FIG. 1A. The baseband processor IC **112A** is an example of the second IC **112** in FIG. 1A.

In the example of FIG. 1B, the baseband processor IC **112A** has a set of serial input channels **114A** and a set of serial output channels **114B**. The set of serial input channels **114A** and the set of serial output channels **114B** are an example of the serial interface **114** in FIG. 1A. As shown, the baseband processor IC **112A** includes a processor **190** and a JESD transceiver **116A**. The JESD transceiver **116A** is an example of the JESD transceiver **116** in FIG. 1A. In the example of FIG. 1B, JESD transceiver **116A** has a set of serial input channels **184**, a set of serial output channels **186**, and a communication interface **188**. The processor **190** also has a communication interface **192**.

The integrated transceiver IC **102A** has an input **156**, an output **158**, a set of serial output channels **104A**, and a set of serial input channels **104B**. The set of serial output channels **104A** and the set of serial input channels **104B** are an example of the serial interface **104** in FIG. 1A. In some examples, the integrated transceiver IC **102A** includes a radio frequency (RF) ADC **160**, a phase-locked loop (PLL) **164**, a RF DAC **168**, a TX digital up-converter (TX DUC) **172**, an RX digital down-converter (RX DDC) **174**, and a JESD transceiver **106A**. The JESD transceiver **106A** is an example of the JESD transceiver **106** in FIG. 1A. The RF ADC **160** has a first input **161**, a second input **162**, and an output **163**. The PLL **164** has a clock input **165**, a first clock output **166**, and a second clock output **167**. The RF DAC **168** has a first input **169**, a second input **170**, and an output **171**. The TX DUC **172** has an input and an output. The RX DDC **174** has an input and an output. The JESD transceiver **106A** has an input **176**, a serial output interface **180**, and a serial input interface **182**.

As shown, the system **130** also includes an RF port **132**, a switch or duplexer **136**, a low-noise amplifier (LNA) **144**, and a power amplifier (PA) **150**. The RF port **132** has a terminal **134** and may include an RF antenna. The switch or duplexer **136** has a terminal **138**, an input **140**, and an output **142**. The LNA **144** has an input and an output. The PA **150** has an input and an output. In some examples, the system **130** includes multiple analog channels, each analog channel having a respective RF port, a respective switch or duplexer, a respective LNA, and a respective PA. In such examples, the integrated transceiver IC **102A** includes additional components such as an RF ADC, an RF DAC, an RX DDC, and a TX DUC for each analog channel. Also, the JESD transceiver **106A** may include inputs (not shown) in addition to the input **176** and outputs (not shown) in addition to the output **178** to support communications to and from each of multiple analog channels.

In the example of FIG. 1B, the terminal **134** of the RF port **132** is coupled to the terminal **138** of the switch or duplexer **136**. The input **140** of the switch or duplexer **136** is coupled to the output of the PA **150**. The input of the PA **150** is coupled to the output **158** of the integrated transceiver IC **102A**. The output **142** of the switch or duplexer **136** is

coupled to the input of the LNA **144**. The output of the LNA **144** is coupled to the input **156** of the integrated transceiver IC **102A**.

The input **156** of the integrated transceiver IC **102A** is coupled to the first input **161** of the RF ADC **160**. The second input **162** of the RF ADC **160** is coupled to the first clock output **166** of the PLL **164**. The output **163** of the RF ADC **160** is coupled to the input of RX DDC **174**. The output of the RX DDC **174** is coupled to the input **176** of the JESD transceiver **106A**. The serial output interface **180** of the JESD transceiver **106A** is coupled to the serial output channels **104A** of the integrated transceiver **102A**. The serial input interface **182** of the JESD transceiver **106A** is coupled to the serial input channels **104B** of the integrated transceiver **102A**. The output **178** of the JESD transceiver **106A** is coupled to the input of the TX DUC **172**. The output of the TX DUC **172** is coupled to the first input **169** of the RF DAC **168**. The second input **170** of the RF DAC **168** is coupled to the second clock output **167** of the PLL **164**. The output **171** of the RF DAC **168** is coupled to the output **158** of the integrated transceiver **102A**.

With the system **130**, data may be transferred from the baseband processor IC **112A** to the RF port **132** via the integrated transceiver **102A**, the PA **150**, and the switch or duplexer **136**. During such data transfers, the processor **190** of the baseband processor IC **112A** may provide data to be transferred. The JESD transceiver **116A** of the baseband processor IC **112A** receives the data provided by the processor **190** via the communications interfaces **192** and **188**; prepares the data for transmission according to a JESD communication protocol; and transmits the prepared data to the JESD transceiver **106A** of the integrated transceiver **102A** via the set of serial output channels **186** of the JESD transceiver **116A**, the set of serial output channels **114B** of the baseband processor IC **112A**, the set of serial input channels **104B** of the integrated transceiver **102A**, and the serial input interface **182** of the JESD transceiver **106A**. In some examples, the JESD transceivers **106A** and **116A** perform flexible link synchronization operations as described herein to complete data transfers from the JESD transceiver **116A** to the JESD transceiver **106A**.

The data received from the JESD transceiver **116A** by the JESD transceiver **106A** is decoded and prepared for transmission as an analog signal by the integrated transceiver **102A**. In some examples, the JESD transceiver **106A** outputs a digital signal at the output **178** responsive to the decoded data. The TX DUC **172** increases the center frequency of the digital signal. Prior to up-converting the digital signal by increasing its center frequency, the TX DUC **172** may optionally interpolate-and-filter the input digital signal.

The RF DAC **168** converts the upconverted digital signal to an analog signal responsive to the clock signal provided at the second clock output **167** of the PLL **164** and received at the second input **170** of the RF DAC **168**. The analog signal is provided at the output **171** of the RF DAC **168** and is conveyed to the input of the PA via the output **158** of the integrated transceiver **102A**. The PA **150** amplifies the analog signal and provides the amplified analog signal at its output. The switch or duplexer **136** receives the amplified analog signal at its input **140**; and provides the amplified analog signal at the terminal **138**, which is coupled to the terminal **134** of the RF port **132**. In some examples, the switch or duplexer **136** conveys analog signals in one direction at a time (to or from the RF port **132**). In other

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examples, the switch or duplexer **136** conveys analog signals in both directions at the same time (to and from the RF port **132**).

In the example of FIG. 1B, data may be transferred from the RF port **132** to the baseband processor IC **112A** via the switch or duplexer **136**, the LNA **144**, and the integrated transceiver **102A**. During such data transfers, the RF port **132** conveys an analog signal to the switch or duplexer **136**. The RF port **132** receives the analog signal at its terminal **138** and provides the analog signal at the output **142**. The LNA **144** receives the analog signal at its input and provides an amplified analog signal at its output. The amplified analog signal is provided from the output of the LNA **144** to the input **156** of the integrated transceiver **102A**.

The amplified analog signal is provided from the input **156** to the first input **161** of the RF ADC **160**. The RF ADC **160** digitizes the received analog signal, resulting in a digital signal at the output **163**. The digital signal provided by the RF ADC **160** has a frequency based on the clock signal provided at the first clock output **166** of the PLL **164** and received at the second input **162** of the RF ADC **160**. The digital signal provided by the RF ADC **160** is received by the RX DDC **174**, which operates to reduce the center frequency of the digital signal. After down-converting the digital signal by reducing its center frequency, the RX DDC **174** may optionally filter-and-decimate the down-converted digital signal.

The JESD transceiver **106A**: receives the down-converted digital signal from the RX DDC **174**; prepares the data for transmission according to a JESD communication protocol; and transmits the prepared data to the JESD transceiver **116A** of the baseband transceiver IC **112A** via the serial output interface **180** of the JESD transceiver **106A**, the set of serial output channels **114B** of the integrated transceiver **102A**, the set of serial input channels **114A** of the baseband processor IC **112A**, and the set of serial input channels **184** of the JESD transceiver **116A**. In some examples, the JESD transceivers **106A** and **116A** perform flexible link synchronization operations as described herein to complete data transfers from the JESD transceiver **106A** to the JESD transceiver **116A**.

The data received from the JESD transceiver **106A** by the JESD transceiver **116A** is decoded and prepared for transmission as a digital signal to the processor **190** via the communication interfaces **188** and **192**. The decoded data received by the processor **190** may be stored, processed, and/or transferred in different examples. With flexible link synchronization operations as described herein, the system **130** improves link synchronization results when bit error rates (BERs) are above a threshold. Such BERs are expected, for example, with JESD 204D.

FIG. 2 is a block diagram showing an example JESD204 system **200**. In some examples, the JESD204 system **200** is a JESD 204D system. As shown, the JESD204 system **200** includes a JESD204 clock generator **202**, a JESD204 transmitter **206**, and a JESD204 receiver **220**. The JESD204 transmitter **206** includes a data generation block **208**, a transport layer **210**, a scrambler **212**, a TX link layer **214**, a physical (PHY) layer **216**, and a TX clock generator **218**. The JESD204 receiver **220** includes a PHY layer **222**, the RX link layer **118**, a de-scrambler **224**, a transport layer **226**, a back-end data processing block **228**, and an RX clock generator **230**. Given the high-speed nature of operations of the JESD204 transmitter **206** and the JESD204 receiver **220**, the related blocks are generally implemented in hardware. In different examples, the data generation block **208** and the

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back-end data processing block **228** may be partitioned between hardware and software, or may be completely implemented in hardware.

In some examples, the JESD204 transmitter **206** is part of the JESD transceiver **106** of FIG. 1A or the JESD transceiver **106A** of FIG. 1B, while the JESD204 receiver **220** is part of the JESD transceiver **116** of FIG. 1A or the JESD transceiver **116A** of FIG. 1B. As another option, the JESD204 transmitter **206** may be part of the JESD transceiver **116** of FIG. 1A or the JESD transceiver **116A** of FIG. 1B, while the JESD204 receiver **220** is part of the JESD transceiver **106** of FIG. 1A or the JESD transceiver **106A** of FIG. 1B.

In the example of FIG. 2, the JESD204 clock generator **202** provides a first device clock (CLK_DEV1) and a transmitter system clock (SYSREF_TX) to the TX clock generator **218** of the JESD204 transmitter **206**. The JESD204 clock generator **202** also provides a second device clock (CLK_DEV2) and a receiver system clock (SYSREF_RX) to the RX clock generator **230** of the JESD204 receiver **220**. In some examples, CLK_DEV1=CLK_DEV2 and SYSREF_TX=SYSREF_RX. In different examples, the TX clock rate and the RX clock rate may be the same or different.

In operation, the data generator block **208** generates transmit data to be transmitted. The transport layer **210** groups converter samples and control bits into frames which are then passed to the TX link layer **214** via the scrambler **212**. The framed transmit data may be provided to scrambler **212**, which performs a data scrambling operation. In some examples, the scrambler **212** is a multiplicative scrambler that multiplies the input signal by a scrambling transfer function. The scrambling operations result in: 1) sufficient transition density; and 2) avoidance of spectral peaks that may be produced when the same data octet repeats from frame to frame.

In such case, the scrambled and mapped transmit data is provided from the scrambler **212** to the TX link layer **214**. The TX link layer **214** operates to: 1) add a synchronization header; 2) perform Reed-Solomon forward error correction (RS-FEC) by adding parity symbols; and 3) output encoded data. The synchronization header facilitates FEC, frame, and lane alignment. The PHY layer **216** receives the encoded data from the TX link layer **214** and transmits the encoded data to the JESD204 receiver **220** as serial data. The TX clock generator **218** operates to generate a frame clock and a local TX alignment clock for the JESD204 transmitter **206** responsive to CLK_DEV1 and SYSREF_TX. As shown, the operations of the transport layer **210** are based on the frame clock provided by the TX clock generator **218**. Also, the operations of the TX link layer **214** and PHY layer **216** are based on the frame clock and the local TX alignment clock provided by the TX clock generator **218**.

The PHY layer **222** of the JESD204 receiver **220** receives the encoded serial data. The RX link layer **118** operates to: 1) align the incoming data using a sync header alignment process; and 2) perform FEC decoding. In some examples, FEC decoding includes correction of errors (if necessary), removal of FEC parity symbols, and removal of the sync header. The de-scrambler **224** operates to de-scramble the received data when the scrambler **212** of the JESD204 transmitter **206** is used. The transport layer **226** ungroups the frames received from the RX link layer **118** into converter samples and control bits. The back-end data processing block **228** performs data processing on the received data. The RX clock generator **230** operates to generate a frame clock and a local RX alignment clock for the JESD204 receiver **220** responsive to CLK_DEV2 and SYSREF_RX.

As shown, the operations of the transport layer **210** are based on the frame clock provided by the RX clock generator **230**. Also, the operations of the RX link layer **118** and PHY layer **222** are based on the frame clock and the local RX alignment clock provided by the RX clock generator **230**.

In the example of FIG. 2, the RX link layer **118** include the flexible link synchronization controller **120**. In some examples, the flexible link synchronization controller **120** is performed as described herein. Without limitation, the flexible link synchronization controller **120** may perform link synchronization based on sync headers inserted into the data stream by TX link layer **214**. Relative to other link synchronization techniques, the mismatch metric and adjustable scaling factor used by the flexible link synchronization controller **120** improves link synchronization when the BER of communications between a transmitter (e.g., the JESD204 transmitter **206**) and a receiver (e.g., the JESD204 receiver **220**) are above a threshold.

FIG. 3 is a block diagram showing an example data preparation process **300**. As shown, the data preparation process **300** includes data mapping at block **302**. The output **304** of block **302** has a rate of R gigabits per second (Gbps). The data rate R depends on the baseband interface rate, the effective number of converters that are mapped on to a single JESD lane, and the number of bits used to represent each sample. For example, if the baseband interface rate (B) is 1.5 gigasymbols per second (Gbps), the number of converters (M) mapped to a single JESD lane is 2, and the number of bits (N) used to represent a sample is 16, then $R=B*M/N$ bits=48 Gbps. Without limitation, R may range from about 30 Gbps to about 100 Gbps. At block **306**, scrambling is performed on the output **304** of block **302**. The output **308** of block **306** has a rate of R Gbps. At block **310**, sync header insertion is performed on the output **308** of block **306**. In some examples, the output **312** of block **310** has a rate of $257/256*R$ Gbps. At block **314**, RS-FEC encoding is performed on the output **312** of block **310**. In some examples, the output **316** of block **314** has a rate of $34/32*R$ Gbps.

In some examples, the data preparation process **300** is performed by a JESD204 transmitter such as the JESD204 transmitter **206** of FIG. 2. In some examples, block **302** is performed by a transport layer such as the transport layer **210** of FIG. 2. In some examples, block **306** is performed by a scrambler such as the scrambler **212** of FIG. 2. In some examples, blocks **310** and **314** are performed by a TX link layer such as the TX link layer **214** of FIG. 2.

FIGS. 4A-4D are block diagrams showing example data structures **404**, **412**, **414**, and **416** related to the data preparation process **300** of FIG. 3. In some examples, a transmitter (e.g., the JESD204 transmitter **206**) and a receiver (e.g., the JESD204 receiver **220**) uses an RS-FEC scheme with different code word sizes and error correction capability, to provide tradeoff between decoding latency and coding gain. In some examples, different RS-FEC modes and related parameters (N_s , K_s , t_s) are possible. N_s is the number of code word symbols, K_s is the number of raw data symbols that are encoded, and t_s is the maximum number of symbol errors that can be corrected in a code word. N_s-K_s is the number of parity symbols. In a first RS-FEC mode: $N_s=544$; $K_s=514$; and $t_s=15$. In a second RS-FEC mode: $N_s=272$; $K_s=258$; and $t_s=7$. In a third RS-FEC mode: $N_s=144$; $K_s=130$; and $t_s=7$. In a fourth RS-FEC mode: $N_s=136$; $K_s=130$; and $t_s=3$. Without limitation, for FIGS. 4A, 4C, and 4D, $N_s=544$ and $K_s=514$ is assumed.

In FIG. 4A, the data structure **404** includes two data groupings **406A** and **406B** (referred to as "code words" in the relevant art). Each of the code words **406A** and **406B**

include 4 multi-blocks. Each of the multi-blocks includes 20 sets of 64-bit blocks. In some examples, the output **304** of the block **302** in FIG. 3 has the data structure **404**.

The data structure **412** in FIG. 4B includes an inserted sync header **414A** for the code word **406A** and an inserted sync header **414B** for the code word **406B**. In some examples, each of the code words **406A** and **406B** has 5120 data bits. Each of the sync headers **414A** and **414B** adds 20 bits for a total of 5140 bits. In some examples, 5140 bits may be represented as 514 symbols (i.e., $K_s=514$). In some examples, the output **312** of the block **310** in FIG. 3 has the data structure **412**. Note: while the scrambling operations of block **306** do not change the data rate, the 5120 bits at the output of block **306** correspond to the 4 multi-blocks used in the data structure **412** of FIG. 4B.

The data structure **414** in FIG. 4C includes 20 sync header bits (SH_{19} to SH_0). In some examples, each of the sync headers **414A** and **414B** of FIG. 4B has the data structure **414**. In some examples, the first 18 bits of a sync header has a fixed pattern. The last 2 bits of the sync header are used to indicate the alignment block boundary, which is the bit location where the extended multi-block boundary (EMB) and RS-FEC code word boundary align. In some examples, the last two bits of the sync header may have a "10" or "01" pattern.

The sync header bits or just "sync bits" are used to determine the alignment between the extended EMB and the RS FEC code word boundary. In some examples, each multi-block of data includes 160 bytes (1280 bits) that are concatenated to create an extended multi-block depending on the transport layer parameters. Example transport layer parameters includes the number of bits/sample, the interface rate versus the lane rate, and/or other parameters. As previously noted, the last 2 bits of the sync header are used to indicate the alignment block boundary, while the first 18 bits of the sync header have a fixed and predetermined pattern. With the last two bits of the sync header limited to "10" or "01", the total number of bits for the sync header is equivalent to 19 bits.

The data structure **416** of FIG. 4D is an RS-FEC encoded bit stream that includes code words and related sync headers separated by parity bits. In some examples, each code word includes 5120 bits and each sync header includes 20 bits. In some examples, the parity bits for each code word and related sync header includes 30 parity symbols. In some examples, the 30 parity symbols include 300 parity bits. In different examples, the number of parity bits, the number of parity bits per symbol, and/or the number of parity symbols may vary. In some examples, the output **316** of the block **314** in FIG. 3 has the data structure **416**. In order to properly decode the data structure **416**, a receiver, such as the JESD204 receiver **220** of FIG. 2, performs link synchronization operations to identify to the location of each sync header.

To successfully decode the encoded RS-FEC bits, the RS-FEC code word boundary needs to be determined by the receiver. To enable this, a sync header pattern is inserted at the beginning of each code word. The data structure **414** of FIG. 4C is an example sync header pattern. The receiver exploits knowledge of the sync header to synchronize to the RS-FEC code word boundary.

At the receiver, the JESD link is synchronized to the RS-FEC code word boundary by searching over N (e.g., $N=5440$ for $N_s=544$) possible bit locations. Example parameters of interest in designing a link synchronization scheme include: synchronization time; false lock probability; and probability of miss. For JESD 204D, robustness to received

bit errors is an additional parameter due to PAM4 having a higher raw BER, when compared to non-return-to-zero (NRZ) encoding/decoding. As used herein “robustness” to received bit errors refers to the ability of the described link synchronization scheme to not fail, even in presence of received bit errors.

FIG. 5 is a block diagram showing an example link synchronization controller 120A. The link synchronization controller 120A is an example of the link synchronization controller 120 in FIGS. 1 and 2. As shown, the link synchronization controller 120A includes a received bit storage unit 502, a received bit set selector 504, a sync pattern match checker 506, a mismatch metric generator 508, a mismatch metric accumulator 510, a link synchronization checker 512, and a sync offset index updater 514. In some examples, the received bit set selector 504, the sync pattern match checker 506, the mismatch metric generator 508, the mismatch metric accumulator 510, the link synchronization checker 512, and the sync offset index updater 514 are implemented as hardware. In other examples, a processor accesses data and executes instructions stored in memory to perform the operations of the received bit set selector 504, the sync pattern match checker 506, the mismatch metric generator 508, the mismatch metric accumulator 510, and the link synchronization checker 512.

The received bit storage unit 502 stores data bits received by the receiver PHY layer such as the PHY layer 222 of FIG. 2. The received bit set selector 504 selects a set of consecutive bits from the received data bits stored by the received bits storage unit 502. In some examples, the received bit set selector 504 selects 20 consecutive bits as a set. The sync pattern match checker 506 compares the selected set of received bits with a predetermined sync header pattern. The mismatch metric generator 508 determines a mismatch metric responsive to results of the comparison performed by the sync pattern match checker. In some examples, the mismatch metric generator 508 determines the mismatch metric responsive to results of the comparison and an adjustable scaling factor.

In some examples, the mismatch metric accumulator 510 determines a cumulative mismatch metric based on iterative operations of the receive bit set selector 504, the sync pattern match checker 506, the mismatch metric generator 508, the link synchronization checker 512, and the sync offset index updater 514. In some examples, the link synchronization checker 512 determines when link synchronization is complete responsive to the cumulative mismatch metric determined by the mismatch metric accumulator 510. In some examples, the link synchronization checker 512 compares the cumulative mismatch metric over multiple iterations to a threshold. If the cumulative mismatch metric is less than the threshold, the link synchronization checker 512 provides a link synchronization status to the sync offset index updater 514, where the link synchronization status indicates link synchronization is successful. In such case, further iterative operations of the sync offset index updater 514, the received bit set selector 504, the sync pattern match checker 506, the mismatch metric generator 508, and the link synchronization checker 512, are terminated or paused. The link synchronization status may also be provided as an output of the link synchronization controller 120A.

If the cumulative mismatch metric is more than the threshold, the link synchronization status indicates link synchronization is not complete. In such case, further iterative operations of the sync offset index updater 514, the received bit set selector 504, the sync pattern match checker

506, the mismatch metric generator 508, and the link synchronization checker 512 continue until link synchronization is successful.

FIG. 6 is a flowchart showing an example link synchronization method 600. The method 600 is performed, for example, by the link synchronization controller 120 of FIGS. 1 and 2, or the link synchronization controller 120A of FIG. 5. As shown, the link synchronization method 600 includes receiving a set of data bits at block 602. At block 604, a comparison of the set of data bits with a sync header pattern is performed. At block 606, a mismatch metric is determined responsive to the comparison and an adjustable scaling factor. The adjustable scaling factor enables comparison mismatches identified at block 604 to have a scaled value that increases or decreases the value of each mismatch. At block 608, link synchronization operations are performed based on the mismatch metric.

In some examples, executing the link synchronization operations at block 608 includes: comparing the mismatch metric to a mismatch threshold; incrementing a sync index offset responsive to the mismatch metric being greater than the mismatch threshold; and repeating the receive, perform, determine, comparing, and incrementing blocks for new sets of data bits based on the sync index offset until a related mismatch metric is less than the mismatch threshold. In some examples, the mismatch metric is a first mismatch metric, and executing the link synchronization operations at block 608 includes: repeating the receive, perform, and determine blocks for X sets of data bits offset by a predetermined offset, where X is an integer greater than 1; determining a cumulative mismatch metric based on the first mismatch metric and each mismatch metric determined for the X sets of data bits; comparing the cumulative mismatch metric to a threshold; and completing link synchronization responsive to the cumulative mismatch metric being less than the threshold.

In some examples, executing the link synchronization operations at block 608 includes: incrementing a sync index offset responsive to the cumulative mismatch metric being greater than the threshold; repeating the receive, perform, and determine blocks for Y new sets of data bits offset by a predetermined offset, where Y is an integer greater than 2; determining a new cumulative mismatch metric based on each mismatch metric determined for the Y sets of data bits; comparing the new cumulative mismatch metric to the threshold; and completing link synchronization responsive to the new cumulative mismatch metric being less than the threshold.

In some examples, the method 600 includes: performing link synchronization based on the adjustable scaling factor having a first value and mismatch metrics for a first number of sets of data bits; and performing link maintenance based on the adjustable scaling factor having a second value and mismatch metrics for a second number of sets of data bits, the second value greater than the first value, and the second number of sets greater than the first number of sets. In some examples, the method 600 includes decoding RS-FEC encoded data based on the completed link synchronization.

FIG. 7 is a flowchart showing another example link synchronization method 700. The method 700 is performed, for example, by the link synchronization controller 120 of FIGS. 1 and 2, or the link synchronization controller 120A of FIG. 5. As shown, the link synchronization method 700 includes initializing a bit separation parameter N, a number of subsequent sets parameter P, and a threshold (TH) at block 702. Also, a count and a cumulative mismatch metric Δ are set to 0 at block 702. At block 704, a set of selected

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received bits corresponding to a current sync index m is compared with a sync header pattern. Also, the count is incremented by 1 at block 704. At block 706, the mismatch metric δ is determined based on the number of bit mismatches from the comparison of block 704 and an adjustable scaling factor. The value of Δ is also incremented by δ at block 706. If Δ is greater than a threshold (block 708), the sync index offset m is incremented by 1, and the count and Δ are set to zero at block 710. The method 700 then returns to block 704. If Δ is not greater than the threshold (block 708) and the count is less than P (block 712), the sync index offset m is incremented by N at block 714. In some examples, N is 5120. After block 714, the method 700 then returns to block 704. If Δ is not greater than the threshold (block 708) and the count is greater than P (block 712), the link synchronization status is complete at block 716.

With the method 700, the number of mismatches that are allowed is flexible using the adjustable scaling factor. Other link synchronization options include: parallel processing of multiple sync index offsets; using a cumulative mismatch metric for one or multiple iterations; and/or performing link maintenance operations. Once link synchronization is complete, the status of the link is continuously monitored to determine loss of synchronization. This is referred to as link maintenance. Loss of synchronization can happen for a variety of reasons. For example, the PHY layer PLL could go out of lock, or there may be cycle slips in the clock and data recovery (CDR) scheme resulting in missed or extra samples. During link maintenance, set of received data bits at index offsets that are integer multiples of N , with respect to the sync index offset determined during link synchronization, are compared against the sync header pattern. A scheme analogous to link synchronization is used, but with potentially different parameters (e.g., different values of TH , P , and/or the scaling factor). The flexible link synchronization operations described herein reduce the probability of missing a valid sync header location due to occurrence of bit errors. Reducing valid sync header misses reduces the overall synchronization time.

FIG. 8 is a diagram showing an example sync pattern match checker 506A and mismatch metric generator 508A. The sync pattern match checker 506A is an example of the sync pattern match checker 506 in FIG. 5. In some examples, the sync pattern match checker 506A includes a set of XOR gates 806. Each XOR gate of the set of XOR gates 806 receives a respective received bit from a set of received bits 802 and a respective bit from a sync header pattern 804. In the example of FIG. 8, the set of received bits 802 includes bits R_m to R_{m+19} . The sync header pattern 804 includes bits SH_{19} to SH_0 . If the respective bits received by an XOR gate have the same value, the XOR gate outputs a logic "0". Otherwise, if the respective bits received by an XOR gate do not have the same value, the XOR gate outputs a logic "1" as a mismatch indicator. The outputs from the XOR gates of the set of XOR gates 806 are coupled to a mismatch metric generator 508A. The mismatch metric generator 508A is an example of the mismatch metric generator 508 in FIG. 5. In operation, the mismatch metric generator 508A generates a mismatch metric δ by adding the mismatch indicators from the set of XOR gates 806 and applying a scaling factor. In some examples, the mismatch metric δ is computed as:

$$\delta = \text{ScaleFactor} * \sum_{i=0}^{19} \{R_{m+i} \oplus SH_{19-i}\} \quad \text{Equation (1)}$$

In some examples, the cumulative mismatch metric for the current Count value is computed by adding the mismatch metric δ determined for the current Count value with the

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cumulative mismatch metric computed previously, i.e., from (Count-1). In other words, $\Delta(\text{Count}) = \Delta(\text{Count}-1) + \delta$ (Count).

FIG. 9 is a block diagram showing an example processing system 900 to perform link synchronization operations. In some examples, the processing system 900 is included with a receiver, such as the JESD204 receiver 220 in FIG. 2, and performs link synchronization operations. In other examples, the processing system 900 is coupled to a receiver, such as the JESD204 receiver 220 in FIG. 2, and performs link synchronization operations. As shown, the processing system 900 includes a processor 902 and a memory 904. The memory 904 stores received data 906 and flexible link synchronization instructions 908. In some examples, the processor 902 executes the flexible link synchronization instructions 908 to perform link synchronization operations. Example link synchronization operations performed by the processor 902, when executing the flexible link synchronization instructions 908, include the operations described for the receive bit set selector 504, the sync pattern match checker 506, the mismatch metric generator 508, the mismatch metric accumulator 510, the link synchronization checker 512, and the sync offset index updater 514. Other example link synchronization operations performed by the processor 902, when executing the flexible link synchronization instructions 908, include the operations of method 600 of FIG. 6 and/or the operations of method 700 of FIG. 7.

In some examples, a receiver, such as the JESD204 receiver 220 of FIG. 2, includes a PHY layer (e.g., the PHY layer 222 in FIG. 2) and a processor (e.g., the processor 902 in FIG. 9). The processor is configured to: receive a set of data bits; perform a comparison of the set of data bits to a sync header pattern; determine a mismatch metric responsive to the comparison and an adjustable scaling factor; and execute link synchronization operations based on the mismatch metric. In some examples, the processor is configured to execute the link synchronization operations by: comparing the mismatch metric to a mismatch threshold; incrementing a sync index offset responsive to the mismatch metric being greater than the mismatch threshold; and repeating the receive, perform, determine, comparing, and incrementing steps for new sets of data bits based on the sync index offset until a related mismatch metric is less than the mismatch threshold. In some examples, the mismatch metric is a first mismatch metric, and the processor is configured to execute the link synchronization operations by: repeating the receive, perform, and determine blocks for X sets of data bits offset by a predetermined offset, where X is an integer greater than 1; determining a cumulative mismatch metric based on the first mismatch metric and each mismatch metric determined for the X sets of data bits; comparing the cumulative mismatch metric to a threshold; and completing link synchronization responsive to the cumulative mismatch metric being less than the threshold.

In some examples, the processor is configured to execute the link synchronization operations by: incrementing a sync index offset responsive to the cumulative mismatch metric being greater than the threshold; repeating the receive, perform, and determine steps for Y new sets of data bits offset by a predetermined offset, where Y is an integer greater than 2; determining a new cumulative mismatch metric based on each mismatch metric determined for the Y sets of data bits; comparing the new cumulative mismatch metric to the threshold; and completing link synchronization responsive to the new cumulative mismatch metric being less than the threshold. In some examples, the processor is

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configured to: perform link synchronization based on the adjustable scaling factor having a first value and mismatch metrics for a first number of sets of data bits; and perform link maintenance based on the adjustable scaling factor having a second value and mismatch metrics for a second number of sets of data bits, the second value greater than the first value, and the second number of sets greater than the first number of sets. In some examples, a receiver includes a link layer (e.g., the RX link layer 118 in FIG. 1) configured to decode RS-FEC encoded data based on the completed link synchronization.

In some examples, a system, such as the system 100, includes: a first IC having a transmitter; and a second IC having a receiver coupled to the transmitter via a serial communication interface. In some examples, the first IC includes an integrated transceiver and the second IC includes a baseband processor.

In some examples, the receiver is configured to: receive a set of data bits; perform a comparison of the set of data bits to a sync header pattern; determine a mismatch metric responsive to the comparison and an adjustable scaling factor; and execute link synchronization operations based on the mismatch metric. In some examples, the receiver is configured to execute the link synchronization operations by: comparing the mismatch metric to a mismatch threshold; incrementing a sync index offset responsive to the mismatch metric being greater than the mismatch threshold; and repeating the receive, perform, determine, comparing, and incrementing steps for new sets of data bits based on the sync index offset until a related mismatch metric is less than the mismatch threshold. In some examples, the mismatch metric is a first mismatch metric, and the receiver is configured to execute the link synchronization operations by: repeating the receive, perform, and determine steps for X sets of data bits offset by a predetermined offset, where X is an integer greater than 1; determining a cumulative mismatch metric based on the first mismatch metric and each mismatch metric determined for the X sets of data bits; comparing the cumulative mismatch metric to a threshold; and completing link synchronization responsive to the cumulative mismatch metric being less than the threshold.

In some examples, the receiver is configured to execute the link synchronization operations by: incrementing a sync index offset responsive to the cumulative mismatch metric being greater than the threshold; repeating the receive, perform, and determine steps for Y new sets of data bits offset by a predetermined offset, where Y is an integer greater than 2; determining a new cumulative mismatch metric based on each mismatch metric determined for the Y sets of data bits; comparing the new cumulative mismatch metric to the threshold; and completing link synchronization responsive to the new cumulative mismatch metric being less than the threshold.

In some examples, the receiver is configured to: perform link synchronization based on the adjustable scaling factor having a first value and mismatch metrics for a first number of sets of data bits; and perform link maintenance based on the adjustable scaling factor having a second value and mismatch metrics for a second number of sets of data bits, the second value greater than the first value, and the second number of sets greater than the first number of sets. In some examples, the receiver includes a link layer configured to decode RS-FEC encoded data based on the completed link synchronization.

In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For

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example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other examples, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incor-

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porated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

What is claimed is:

1. A receiver comprising:
 - a PHY layer; and
 - a processor coupled to the PHY layer and configured to:
 - receive a set of data bits from the PHY layer;
 - compare the set of data bits to a sync header pattern, wherein the comparing includes comparing each bit of the set of data bits to a respective bit of the sync header pattern to detect a corresponding bit mismatch;
 - determine a mismatch metric based on the corresponding bit mismatches and an adjustable scaling factor; and
 - execute link synchronization operations based on the mismatch metric.
2. The receiver of claim 1, wherein the processor is configured to execute the link synchronization operations by:
 - comparing the mismatch metric to a mismatch threshold;
 - incrementing a sync index offset responsive to the mismatch metric being greater than the mismatch threshold; and
 - repeating the receive, determine, comparing, and incrementing for new sets of data bits based on the sync index offset until a related mismatch metric is less than the mismatch threshold.
3. The receiver of claim 1, wherein the mismatch metric is a first mismatch metric, and the processor is configured to execute the link synchronization operations by:
 - repeating the receive, and determine for X sets of data bits offset by an offset, where X is an integer greater than 1;
 - determining a cumulative mismatch metric based on the first mismatch metric and each mismatch metric determined for the X sets of data bits;
 - comparing the cumulative mismatch metric to a threshold; and
 - completing link synchronization responsive to the cumulative mismatch metric being less than the threshold.
4. The receiver of claim 3, wherein the processor is configured to execute the link synchronization operations by:
 - incrementing a sync index offset responsive to the cumulative mismatch metric being greater than the threshold;
 - repeating the receive, and determine for Y new sets of data bits offset by an offset, where Y is an integer greater than 2;
 - determining a new cumulative mismatch metric based on each mismatch metric determined for the Y sets of data bits;
 - comparing the new cumulative mismatch metric to the threshold; and

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completing link synchronization responsive to the new cumulative mismatch metric being less than the threshold.

5. The receiver of claim 3, further comprising a link layer configured to decode Reed-Solomon Forward Error Correction (RS-FEC) encoded data based on the completed link synchronization.

6. The receiver of claim 1, wherein the processor is configured to:

- perform link synchronization based on the adjustable scaling factor having a first value and mismatch metrics for a first number of sets of data bits; and
- perform link maintenance based on the adjustable scaling factor having a second value and mismatch metrics for a second number of sets of data bits, the second value greater than the first value, and the second number of sets greater than the first number of sets.

7. A method comprising:

- receiving a set of data bits;
- compare the set of data bits to a sync header pattern, wherein the comparing includes comparing each bit of the set of data bits to a respective bit of the sync header pattern to detect a corresponding bit mismatch;
- determine a mismatch metric based on the corresponding bit mismatches and an adjustable scaling factor; and
- execute link synchronization operations based on the mismatch metric.

8. The method of claim 7, wherein executing the link synchronization operations includes:

- comparing the mismatch metric to a mismatch threshold;
- incrementing a sync index offset responsive to the mismatch metric being greater than the mismatch threshold; and
- repeating the receive, determine, comparing, and incrementing for new sets of data bits based on the sync index offset until a related mismatch metric is less than the mismatch threshold.

9. The method of claim 7, wherein the mismatch metric is a first mismatch metric, and executing the link synchronization operations includes:

- repeating the receive, and determine for X sets of data bits offset by an offset, where X is an integer greater than 1;
- determining a cumulative mismatch metric based on the first mismatch metric and each mismatch metric determined for the X sets of data bits;
- comparing the cumulative mismatch metric to a threshold; and
- completing link synchronization responsive to the cumulative mismatch metric being less than the threshold.

10. The method of claim 9, wherein executing the link synchronization operations includes:

- incrementing a sync index offset responsive to the cumulative mismatch metric being greater than the threshold;
- repeating the receive, and determine for Y new sets of data bits offset by an offset, where Y is an integer greater than 2;
- determining a new cumulative mismatch metric based on each mismatch metric determined for the Y sets of data bits;
- comparing the new cumulative mismatch metric to the threshold; and
- completing link synchronization responsive to the new cumulative mismatch metric being less than the threshold.

11. The method of claim 9, further comprising decoding Reed-Solomon Forward Error Correction (RS-FEC) encoded data based on the completed link synchronization.

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12. The method of claim 7, further comprising:
 performing link synchronization based on the adjustable
 scaling factor having a first value and mismatch metrics
 for a first number of sets of data bits; and
 performing link maintenance based on the adjustable
 scaling factor having a second value and mismatch
 metrics for a second number of sets of data bits, the
 second value greater than the first value, and the second
 number of sets greater than the first number of sets.

13. A system comprising:
 a first integrated circuit (IC) having a transmitter;
 a serial communication interface; and
 a second IC having a receiver in communication with the
 transmitter via the serial communication interface, the
 receiver configured to:
 receive a set of data bits;
 compare the set of data bits to a sync header pattern,
 wherein the comparing includes comparing each bit
 of the set of data bits to a respective bit of the sync
 header pattern to detect a corresponding bit mis-
 match;
 determine a mismatch metric based on the correspond-
 ing bit mismatches and an adjustable scaling factor;
 and
 execute link synchronization operations based on the
 mismatch metric.

14. The system of claim 13, wherein the first IC includes
 a baseband processor and the second IC includes a data
 converter.

15. The system of claim 13, wherein the first IC includes
 a data converter and the second IC includes a baseband
 processor.

16. The system of claim 13, wherein the receiver is
 configured to execute the link synchronization operations
 by:
 comparing the mismatch metric to a mismatch threshold;
 incrementing a sync index offset responsive to the mis-
 match metric being greater than the mismatch thresh-
 old; and
 repeating the receive, determine, comparing, and incre-
 menting for new sets of data bits based on the sync
 index offset until a related mismatch metric is less than
 the mismatch threshold.

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17. The system of claim 16, wherein the receiver includes
 a link layer configured to decode Reed-Solomon Forward
 Error Correction (RS-FEC) encoded data based on a com-
 pleted link synchronization.

18. The system of claim 13, wherein the mismatch metric
 is a first mismatch metric, and the receiver is configured to
 execute the link synchronization operations by:
 repeating the receive, and determine for X sets of data bits
 offset by an offset, where X is an integer greater than 1;
 determining a cumulative mismatch metric based on the
 first mismatch metric and each mismatch metric deter-
 mined for the X sets of data bits;
 comparing the cumulative mismatch metric to a threshold;
 and
 completing link synchronization responsive to the cumu-
 lative mismatch metric being less than the threshold.

19. The system of claim 18, wherein the receiver is
 configured to execute the link synchronization operations
 by:
 incrementing a sync index offset responsive to the cumu-
 lative mismatch metric being greater than the threshold;
 repeating the receive, and determine for Y new sets of data
 bits offset by an offset, where Y is an integer greater
 than 2;
 determining a new cumulative mismatch metric based on
 each mismatch metric determined for the Y sets of data
 bits;
 comparing the new cumulative mismatch metric to the
 threshold; and
 completing link synchronization responsive to the new
 cumulative mismatch metric being less than the thresh-
 old.

20. The system of claim 13, wherein the receiver is
 configured to:
 perform link synchronization based on the adjustable
 scaling factor having a first value and mismatch metrics
 for a first number of sets of data bits; and
 perform link maintenance based on the adjustable scaling
 factor having a second value and mismatch metrics for
 a second number of sets of data bits, the second value
 greater than the first value, and the second number of
 sets greater than the first number of sets.

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